NexusOS
AI Business Operating System

Part 03.0
Linear Issue WAL-11

Schema Implementation

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-11
Report title	Schema Implementation
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Purpose	Standalone Linear attachment (manual upload)
Source report(s)	docs/reports/Part-03.0/Part-03.0-Schema-Implementation.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

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Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Schema Implementation

Linear: WAL-11 **Reconstruction date:** 2026-07-19

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Objective

Implement the enterprise schema in SQLAlchemy models aligned with the approved multi-tenant model.

What was implemented

Model package (`backend/app/models/`)

At Part 03.0 landing (`838db64` `__init__.py`), registered models included Company, User/Role/Permission (+ association tables), CRM, Project/Task, HR, Finance, Inventory, Document, Notification, AuditLog, AI, Workflow, Activity.

Note: `Session` / `AuthToken` *models were added later by Part 02 (migration* `382d8497298e`*) and appear in the current* `__init__.py` *— they are* ***not*** *Part 03.0 deliverables.*

Mixins and base (`base.py`)

- `Base` with UUID v7 PK + naming convention metadata
- `TenantMixin`, `TimestampMixin`, `SoftDeleteMixin`, `AuditedByMixin`, `ActiveMixin`
- `TenantBusinessMixin` composition

Repository foundation

`BaseRepository` with tenant-scoped `_base_query()`, soft-delete exclusion, and fail-closed missing `company_id`.

Later Part 03.1 substantially expanded repository methods (pagination clamps, mass-assignment protection, etc.). Those expansions are Part 03.1 evidence, not Part 03.0.

Seed

`backend/app/db/seed.py` — 53 permissions from `MODULES`; default role permission patterns.

Repository evidence

- `backend/app/models/**`
- `backend/app/repositories/base.py` (core tenant filter at `838db64`; later hardened)
- `backend/app/db/seed.py`
- `backend/app/utils/uuid7.py`
- Commit `838db64`

Validation

Models register with `Base.metadata` for Alembic; initial migration creates 37 tables; isolation tests exercise repository behavior on tenant models.

Outcome

Enterprise schema implemented in ORM form, ready for migration and authentication to attach.

Evidence limitations (shared)

- Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.
- Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.
- No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).
- Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.